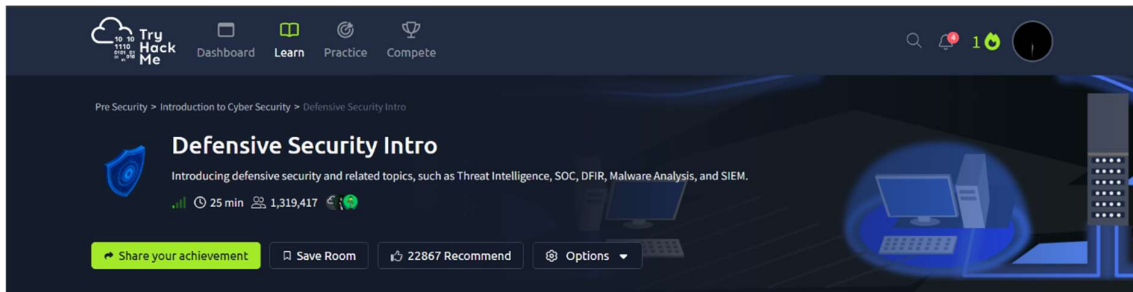


Day 01: Defensive Security Intro



Link: <https://tryhackme.com/room/defensivesecurityintro>

Task 01: Introduction to Defensive Security

Task 1 Introduction to Defensive Security

In the previous lesson, we learned about offensive security, which aims to identify and exploit system vulnerabilities to enhance security measures. This includes exploiting software bugs, leveraging insecure setups, and taking advantage of unenforced access control policies, among other strategies. Red teams and penetration testers specialize in these offensive techniques.

In this lesson, we will examine its counterpart, defensive security. It is concerned with two main tasks:

1. Preventing intrusions from occurring
2. Detecting intrusions when they occur and responding properly

Blue teams are part of the defensive security landscape.

Some of the tasks that are related to defensive security include:

- User cyber security awareness: Training users about cyber security helps protect against attacks targeting their systems.
- Documenting and managing assets: We need to know the systems and devices we must manage and protect adequately.
- Updating and patching systems: Ensuring that computers, servers, and network devices are correctly updated and patched against any known vulnerability (weakness).
- Setting up preventative security devices: firewall and intrusion prevention systems (IPS) are critical components of preventative security. Firewalls control what network traffic can go inside and what can leave the system or network. IPS blocks any network traffic that matches present rules and attack signatures.
- Setting up logging and monitoring devices: Proper network logging and monitoring are essential for detecting malicious activities and intrusions. If a new unauthorized device appears on our network, we should be able to detect it.

There is much more to defensive security. Aside from the above, we will also cover the following related topics:

- Security Operations Center (SOC)
- Threat Intelligence
- Digital Forensics and Incident Response (DFIR)
- Malware Analysis

Answer the questions below

Which team focuses on defensive security?

Blue Team

✓ Correct Answer

Task 02: Areas of Defensive Security

Task 2 Areas of Defensive Security

In this task, we will cover two main topics related to defensive security:

- Security Operations Center (SOC), where we cover Threat Intelligence
- Digital Forensics and Incident Response (DFIR), where we also cover Malware Analysis

Security Operations Center (SOC)

A *Security Operations Center (SOC)* is a team of cyber security professionals that monitors the network and its systems to detect malicious cyber security events. Some of the main areas of interest for a SOC are:

- Vulnerabilities: Whenever a system vulnerability (weakness) is discovered, it is essential to fix it by installing a proper update or patch. When a fix is unavailable, the necessary measures should be taken to prevent an attacker from exploiting it. Although remediating vulnerabilities is vital to a SOC, it is not necessarily assigned to them.
- Policy violations: A security policy is a set of rules required to protect the network and systems. For example, it might be a policy violation if users upload confidential company data to an online storage service.
- Unauthorized activity: Consider the case where a user's login name and password are stolen, and the attacker uses them to log into the network. A SOC must detect and block such an event as soon as possible before further damage is done.
- Network intrusions: No matter how good your security is, there is always a chance for an intrusion. An intrusion can occur when a user clicks on a malicious link or when an attacker exploits a public server. Either way, when an intrusion occurs, we must detect it as soon as possible to prevent further damage.

Security operations cover various tasks to ensure protection; one such task is threat intelligence.

Answer the questions below

What would you call a team of cyber security professionals that monitors a network and its systems for malicious events?

✓ Correct Answer Hint

What does DFIR stand for?

✓ Correct Answer

Which kind of malware requires the user to pay money to regain access to their files?

✓ Correct Answer

Task 03: Practical Example of Defensive Security

Task 3 Practical Example of Defensive Security

The Scenario

Let us pretend you are a *Security Operations Center (SOC)* analyst responsible for protecting a bank. This bank's SOC uses a *Security Information and Event Management (SIEM)* tool, which gathers security-related information and events from various sources and presents them in one dashboard. If the SIEM finds something suspicious, an alert will be generated.



Simulating a SIEM

We have prepared a simplified, interactive simulation of a SIEM system to provide you with a hands-on experience similar to what cyber security analysts encounter.

To start this simulation, please click the "View Site" button below.

 [View Site](#)

This action will open a "static site" on the right side of your screen. Follow the step-by-step instructions provided within the simulation to navigate through the events and locate the "flag." A flag is a series of characters with a format like this: "THM{RANDOM_WORDS}". Use this flag to answer questions from lessons here in TryHackMe, like the one below.

What's next?

In this lesson, we've discussed the different subfields (SOC, Threat Intelligence, Malware Analysis, and DFIR) and experienced firsthand how to deal with alerts in a simulated SIEM environment. While we've covered a lot, the depth and complexity of this field mean there's more to learn and explore. The lessons learned here will serve as your foundation as cyber threats evolve, demanding continuous learning, vigilance, and adaptation.

Continue learning by checking out the next lesson in this series, "Search Skills." This lesson will teach you valuable techniques for searching for information online to aid your investigations and learning.

Answer the questions below

What is the flag that you obtained by following along?

THM{THREAT-BLOCKED}

✓ Correct Answer



You did it! 🎉 Defensive Security Intro complete!

Points earned 🔥 40	Completed tasks 📋 3	Room type 👤 Walkthrough	Difficulty 📶 Easy	Streak 🔥 2
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